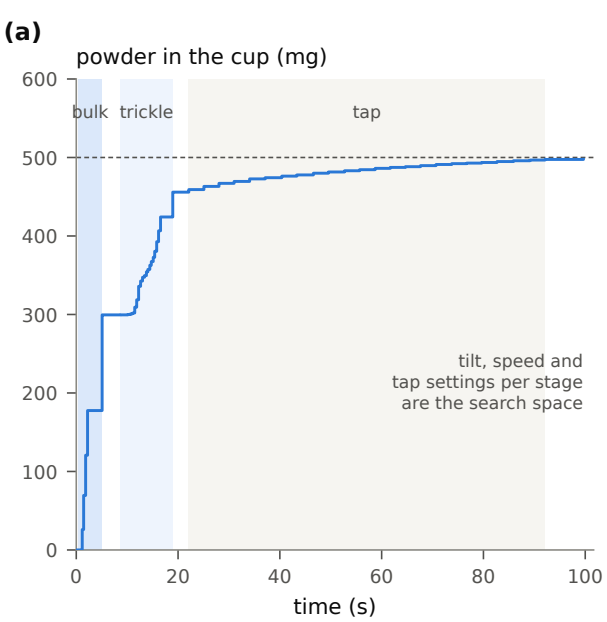
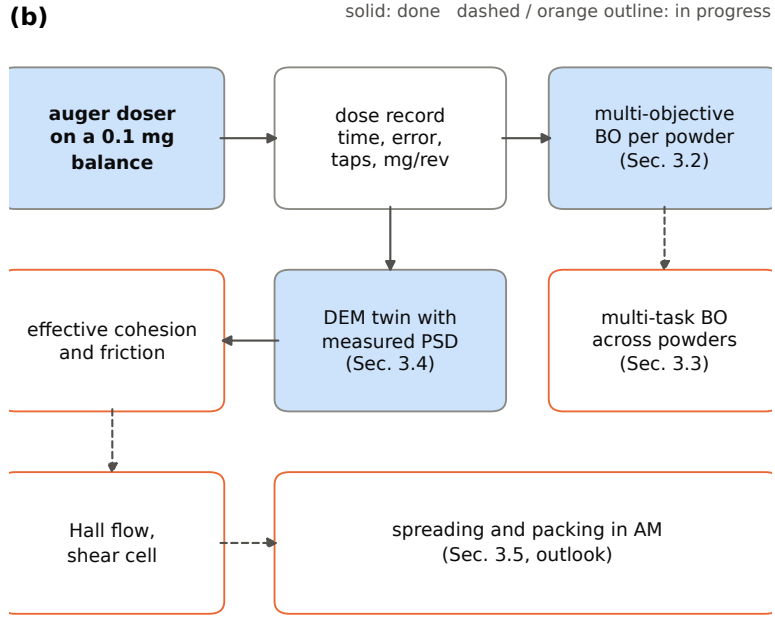

(a)
powder in the cup (mg)
600
500
400
300
200
100
0
bulk
trickle
tap
tilt, speed and
tap settings per stage
are the search space
0
20
40
60
80
100
time (s)


(b)
solid: done dashed / orange outline: in progress
auger doser on a 0.1 mg balance
dose record time, error, taps, mg/rev
multi-objective BO per powder (Sec. 3.2)
effective cohesion and friction
DEM twin with measured PSD (Sec. 3.4)
multi-task BO across powders (Sec. 3.3)
Hall flow, shear cell
spreading and packing in AM (Sec. 3.5, outlook)